PDF Report Enhancement V2 - Professional Electrical Assessment Report

Objective

Transform the current PDF from a simple calculation summary into a professional Electrical Assessment Report.

The PDF should feel like it was generated by an electrical consultant rather than a calculator.

Users should be comfortable:

- Saving it
- Sharing it
- Sending it to electricians
- Using it for planning decisions

The PDF must become a premium output of the platform.

Current Problems

Current PDF contains:

- Summary information
- Recommendations
- Health score
- Savings opportunities

However it lacks:

- Executive summary
- Assessment verdict
- Recommendation reasoning
- Visual hierarchy
- Engineering report structure

The website currently feels more advanced than the PDF.

The PDF should match or exceed the quality of the Assessment page.

Design Principles

The report should follow this structure:

Cover Page

↓

Executive Summary

↓

Electrical Assessment Verdict

↓

Load Analysis

↓

Recommendation Matrix

↓

Detailed Recommendations

↓

Consumption Analysis

↓

Health Assessment

↓

Energy Optimization

↓

Action Plan

↓

Disclaimer

SECTION 1 - Cover Page

Create a dedicated cover page.

Display:

Electrical Assessment Report

Prepared by

ElectricityLoadCalculator.com

Generated Date

Assessment ID

(optional)

Include a professional subtitle:

Electrical Load Analysis, Safety Recommendations, Energy Assessment and Planning Report

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SECTION 2 - Executive Summary

This should be the first section after the cover page.

Purpose:

Allow users to understand the entire report within 10 seconds.

Display large KPI cards:

Connected Load

Daily Consumption

Monthly Consumption

Health Score

Top Consumer

Assessment Status

Example:

Connected Load

5.02 kW

Monthly Consumption

108 kWh

Health Score

70/100

Top Consumer

Refrigerator

Assessment Status

GOOD

SECTION 3 - Electrical Assessment Verdict

Add a dedicated verdict card.

Example:

Electrical Assessment Verdict

Overall Status: Good

Key Findings:

✓ Electrical load is within normal residential range

✓ Existing electrical sizing recommendations are suitable

✓ Solar adoption is feasible

✓ Inverter backup planning is recommended

⚠ Refrigerator contributes significant energy usage

⚠ Energy optimization opportunities identified

This should feel like an executive summary from a consultant.

SECTION 4 - Load Analysis

Display:

Connected Load

Daily Consumption

Monthly Consumption

Operating Current

Power Factor Assumptions (if available)

Include a visual summary box.

Example:

Load: 5.02 kW

Voltage: 230V

Operating Current: 21.8A

SECTION 5 - Recommendation Matrix

Create a dedicated table.

Example:

Recommendation	Value
MCB	32A
Cable	6 sq mm
Inverter	6 kVA
Solar	1.0 kW

This becomes the quick-reference section.

Users should be able to find all recommendations in one place.

SECTION 6 - Detailed Recommendations

Create individual recommendation sections.

MCB Recommendation

Display:

Recommended MCB

32A

Why Recommended

Connected Load

5.02 kW

↓

Current

21.8 A

↓

Safety Margin

25%

↓

Recommended MCB

32A

Explain:

Why not smaller?

Why not larger?

Cable Recommendation

Display:

Recommended Cable

6 sq mm Copper

Calculation Path

Current

21.8 A

↓

MCB

32A

↓

Recommended Cable

6 sq mm

Explain cable selection reasoning.

Inverter Recommendation

Display:

Recommended Inverter

6 kVA

Calculation Path

Load

5.02 kW

↓

Sizing Factor

1.2

↓

Required Capacity

6.02 kVA

↓

Recommended Inverter

6 kVA

Solar Recommendation

Display:

Recommended Solar

1.0 kW

Calculation Path

Monthly Units

108

↓

Solar Production Assumption

↓

Required Solar Capacity

↓

Recommended System Size

SECTION 7 - Consumption Analysis

Move existing charts into dedicated section.

Display:

Consumption Breakdown

Top Consumer

Highest Category

Distribution Chart

Add:

Key Observation

Example:

Refrigeration contributes 49% of total energy consumption.

This represents the largest energy optimization opportunity.

SECTION 8 - Electrical Health Assessment

Expand Health Score.

Current score:

70/100

Users need to know why.

Display:

Health Score Breakdown

Load Balance

Energy Efficiency

Safety Readiness

Solar Readiness

Future Expansion Readiness

Example:

Load Balance

18/25

Energy Efficiency

15/25

Safety

20/20

Solar Readiness

17/20

Expansion Readiness

10/10

Total

70/100

Add explanations for deductions.

SECTION 9 - Energy Optimization Opportunities

Keep existing recommendations.

Improve formatting.

Display:

Potential Savings

Priority

Estimated Impact

Categorize:

High Priority

Medium Priority

Low Priority

SECTION 10 - Action Plan

Add a consultant-style action plan.

Example:

Immediate Actions

✓ Replace inefficient refrigerator

✓ Review appliance usage patterns

Short-Term Actions

✓ Consider inverter sizing

✓ Improve energy efficiency

Long-Term Actions

✓ Evaluate rooftop solar installation

✓ Monitor monthly energy consumption

This makes the report actionable.

SECTION 11 - Report Footer

Remove all localhost references.

Never display development URLs.

Always display:

https://electricityloadcalculator.com

Display:

Generated by ElectricityLoadCalculator.com

Electrical Assessment Platform

Visual Improvements

Use:

Large section headers

Clear spacing

Divider lines

Professional card layouts

Consistent typography

Readable tables

Page numbering

PDF UX Requirements

Users should be able to:

- Understand results in under 30 seconds
- Print the report cleanly
- Share with electricians
- Use as planning documentation

Success Criteria

The PDF should no longer feel like:

"Calculator Output"

It should feel like:

"Professional Electrical Assessment Report"

A user viewing only the PDF should immediately understand:

✓ Their electrical load

✓ Their consumption profile

✓ Their health score

✓ Their recommendations

✓ Why those recommendations were generated

✓ What actions they should take next

The PDF should become the most professional asset produced by the platform and reinforce ElectricityLoadCalculator.com as an Electrical Consultant Platform rather than a simple calculator.